

Common Perceptual and Cognitive Biases

Perceptual Biases

Expectations. We tend to perceive what we expect to perceive. More (unambiguous) information is needed to recognize an unexpected phenomenon.

Resistance. Perceptions resist change even in the face of new evidence.

Ambiguities. Initial exposure to ambiguous or blurred stimuli interferes with accurate perception, even after more and better information becomes available.

Biases in Evaluating Evidence

Consistency. Conclusions drawn from a small body of consistent data engender more confidence than ones drawn from a larger body of less consistent data.

Missing Information. It is difficult to judge well the potential impact of missing evidence, even if the information gap is known.

Discredited Evidence. Even though evidence supporting a perception may be proved wrong, the perception may not quickly change.

Biases in Estimating Probabilities

Availability. Probability estimates are influenced by how easily one can imagine an event or recall similar instances.

Anchoring. Probability estimates are adjusted only incrementally in response to new information or further analysis.

Overconfidence. In translating feelings of certainty into a probability estimate, people are often overconfident, especially if they have considerable expertise.

Biases in Perceiving Causality

Rationality. Events are seen as part of an orderly, causal pattern. Randomness, accident and error tend to be rejected as explanations for observed events. For example, the extent to which other people or countries pursue a coherent, rational, goal-maximizing policy is overestimated.

Attribution. Behavior of others is attributed to some fixed nature of the person or country, while our own behavior is attributed to the situation in which we find ourselves.

Intelligence analysts must actively review the accuracy of their mind-sets by applying structured analytic techniques that will make those mental models more explicit and expose their key assumptions. The techniques found in this primer are designed to assist in this regard by:

- Instilling more structure into the intelligence analysis.
- Making analytic arguments more transparent by articulating them and challenging key assumptions.
- Stimulating more creative, “out-of-the-box” thinking and examining alternative outcomes, even those with low probability, to see if available data might support these outcomes.

- Identifying indicators of change (or signposts) that can reduce the chances of surprise.

Incorporating findings derived from these techniques into our intelligence products also serves the policymaker by:

- Highlighting potential changes that would alter key assessments or predictions.
- Identifying key assumptions, uncertainties, intelligence gaps and disagreements that might illuminate risks and costs associated with policy choices.
- Exploring alternative outcomes for which policy actions might be necessary.